

Fruit Fly

Team: Cheesecake

Element H: Prototype Testing and Data Collection Plan

Our boarding school residents need a way to remove fruit flies from their dormitories. This is because residents are irritated by the fruit flies and they could also pose a health risk. The fruit flies are attracted to a food item that was left in the dormitories a few weeks ago. Fruit flies have a short reproduction time so the quicker they are removed the easier it will be because there are less flies to remove. Quality of life in the dormitories would also improve if the fruit flies were to be removed.

Our prototype should have no odor, have a maximum size of 1,000 cm³, maximum cost of \$20, remove 80% of target flies, and material strength: will not break if touched, moved, or dropped. This will give us information on how well the prototype meets our design requirements.

To test for odor:

1. Install prototype and fill with vinegar.
2. Waft over the prototype at a distance of 6 inches.
3. Record observations.
4. Have four additional people repeat steps 1-3.

To test size:

1. Measure height, width, and length of the prototype in centimeters (cm).
2. Record the volume.

To test cost:

1. Add the cost of all materials.

2. Record the total cost.

To test how many flies have been removed:

1. Install prototype and fill with vinegar.
2. Record how many flies are in the target area.
3. Place the prototype in the target area for 24 hours
4. Record how many flies are in the prototype.
5. Repeat steps 1-4, four additional times.

To test the material strength:

1. Drop prototype from table height.
2. Record if the prototype is broken or not.
3. Repeat steps 1-2, four additional times.

Our expert, Ms. Z, reviewed our testing plan and provided feedback. The updated testing plan had improved clarity by making the steps easier to read and follow. Also, our expert helped us create a testing plan that thoughtfully tests for what we want and also includes repetition for more accurate results.